DEEPAK YADAV

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PROFESSIONAL PROFILE

Enthusiastic and detail-oriented Computer Science Graduate with a strong foundation in **Python, SQL, AI/ML models, and Data Engineering** fundamentals. Hands-on experience developing **LLM applications using LangChain, RAG pipelines, and Vector DBs** alongside building robust ETL pipelines. Eager to contribute to enterprise-grade Data & AI products at IBM Bengaluru through strong problem-solving and software development skills.

EDUCATION

Master of Computer Application (MCA) 2024 – 2026
Major in Software & Data Engineering | **CGPA: 8.05/10.0**

8.05 / 10	3+ Projects	Top 5%
MCA CGPA with Specialization in Data & AI	Production-Ready GenAI & Data Engineering Tools	National Level AI/ML Hackathon Finalist

TECHNICAL SKILLS

Programming Languages: Python (Proficient), SQL, Java, C++

Data Engineering & DBs: PySpark, PostgreSQL, MySQL, Vector DBs (ChromaDB, FAISS)

Developer Tools: Git, GitHub, Docker, REST APIs, FastAPI, Airflow (Basic)

AI & Generative AI: LLMs, LangChain, RAG Architecture, OpenAI API, Hugging Face

ML & Data Science: Scikit-Learn, Pandas, NumPy, Matplotlib, Data Wrangling

Core Concepts: Data Structures & Algorithms, OOP, Database Systems (DBMS), OS

ACADEMIC & CAPSTONE PROJECTS

Multi-Document RAG Q&A Assistant with LLM Python, LangChain, ChromaDB, Streamlit

- Built an end-to-end Retrieval-Augmented Generation (RAG) agent that parses complex PDF documents and generates context-aware answers using OpenAI LLM endpoints.
- Implemented vector embeddings with ChromaDB and semantic search, achieving sub-second retrieval latency for queries.

Real-Time Data Pipeline & Analytics Dashboard Python, PySpark, PostgreSQL, SQL, Docker

- Designed an automated ETL data pipeline using **PySpark and PostgreSQL** to aggregate and clean over 100k daily mock transactions.
- Engineered optimized SQL queries and database indexes to accelerate analytical reporting response times by 35%.

INTERNSHIP EXPERIENCE

Software / Data Engineering Intern | **Innovate Tech Labs** Bengaluru, KA | Jun 2025 – Aug 2025

- Developed backend RESTful microservices using **FastAPI and Python** for real-time model inference.
- Wrote and executed automated unit tests and benchmarked SQL database queries for enhanced pipeline stability.
- Collaborated in an Agile team setup with daily standups, code reviews, and Git version control.

CERTIFICATIONS & ACHIEVEMENTS

- **IBM Certified Data Science Professional** & **DeepLearning.AI Generative AI with LLMs**
- **Hackathon Finalist:** Ranked in Top 5% among 200+ teams in National AI Innovation Challenge (2025).
- **Problem Solving:** Solved 250+ DSA problems across LeetCode and HackerRank.